Ghislain THEUBO

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Professional Summary

Embedded Software Engineer (Ensimag) focused on **real-time firmware** and human–machine control loops. At **INRIA**, built C/FreeRTOS modules for an assisted wheelchair: inputs, latency-sensitive pipelines, and continuous validation with the operator. Strong C/C++, MCU/RTOS notions (SPI/I2C/UART), Python tooling, and hardware/software collaboration. Seeking **Embedded Software Engineer – GCS** at Harmattan AI (Paris) to develop firmware for handheld UAV ground stations — inputs, radio link interface, power/responsiveness, and field stability.

Core Strengths

C/C++ firmware · **FreeRTOS / real-time** · Operator control loops · Serial & radio-link notions · Power/latency trade-offs · HW/SW co-debug · Clean production code · English

Experience

Shift Technology – Platform Software Engineering — Paris, France
Software Engineer (CDI – Platform / Backend) — *Sept 2026 – Nov 2026*

- Contributed to the **platform / foundations** squad building shared services used by product teams: design and implementation of **C# / .NET** components and **high-performance APIs** on **Microsoft Azure**.
- Worked with cloud data and messaging building blocks (**Azure Cosmos DB**, **Service Bus**), configuration/transversal services, and infrastructure-as-code awareness (**Terraform**) in a hybrid SaaS environment.
- Practiced production engineering habits: design discussions, code reviews, monitoring mindset, incremental delivery, and collaboration with platform and product engineers.

Gustave Roussy & CentraleSupélec MICS Lab — Paris / Saclay, France
Software Engineer (CDD – Backend / Data Platforms) — *Jan 2026 – Sept 2026*

- Python/FastAPI production APIs with validation, RBAC, logging; PostgreSQL indexing (**∼30%** faster critical reads); Docker + CI.
- Sat with domain users; turned ambiguous needs into reliable software with clear operational docs.

Dassault Systèmes – 3DEXPERIENCE / 3DSwym R&D — Vélizy-Villacoublay, France
Software Engineer Intern (Full-Stack Module Rewrite) — *Mar 2025 – Sept 2025*

- Survey module rewrite in Java 17 / Spring Boot / Hibernate with TDD; **∼35%** perf, **∼85%** coverage — industrial R&D quality bar.

INRIA Grenoble – Humanlab / Froll'N'Roll — Montbonnot-Saint-Martin, France
Research Intern – Real-Time Control (Assistive Wheelchair) — *May 2024 – Aug 2024*

- **C/FreeRTOS** control modules for an operator-driven assistive device — direct analogy to GCS handheld control: inputs, responsiveness, safety under latency.
- Redesigned communication pipelines; **∼40%** responsiveness gain; continuous field validation with the end user.
- DIY HW/SW integration with electrical/mechanical partners; iterative bring-up and debugging.

LIG – Laboratoire d'Informatique de Grenoble — Grenoble, France
Cloud Infrastructure Intern (OpenStack / Grid'5000) — *Feb 2025 – Mar 2025*

- Automation scripts and measurement under constrained environments — discipline transferable to firmware tooling.

Selected Projects

Handheld Control Firmware Sketch | C, FreeRTOS, UART, input debounce — GitHub

- MCU-oriented tasks for joystick/button sampling, debounce, and command framing over UART — GCS-like control path.
- Power/responsiveness notes: sleep modes vs poll rates; logging for field debug.

INRIA Operator Control Loop | C, FreeRTOS, Python — INRIA

- Production-minded real-time control with human-in-the-loop validation — closest proof for GCS operator experience.

Radio Link Framing Lab | C++, CRC, packet loss simulation — GitHub

- Simple framed packets with CRC and drop/reorder stress tests — warm-up for UAV radio interface work.

Technical Skills

Firmware: C, C++, FreeRTOS, MCU notions (ARM plus), HAL, SPI/I2C/UART
Systems: Embedded Linux notions, multi-thread, latency & power trade-offs
Tooling: Git, Python scripts, unit tests, bring-up discipline
Fit: operator-facing control + HW collaboration (INRIA)

Education & Certifications

Grenoble INP - Ensimag — Grenoble, France
Engineering Degree – Information Systems / Software Engineering (Double Degree) — *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé — Yaoundé, Cameroon
Master's Degree in Computer Engineering – Top 12% of class — *2019 – 2023*

Certifications: Programming Foundations – TDD (LinkedIn, 2025) · Primerli – Insurance Industry Foundations (2025)

Languages & Extra

English: bilingual / professional · **French:** fluent · **Spanish:** intermediate
Extra: tutoring · tennis, football